Question		Answer	Marks	AO	Guidance
12		**DR**			
		$3(1+\tan^2\theta)+2\tan\theta-4[=0]$	**B1**	**3.1a**	correct substitution of $\sec^2\theta = 1+\tan^2\theta$ in equation; may be implied by $3+3\tan^2\theta+2\tan\theta-4[=0]$
		$3\tan^2\theta+2\tan\theta-1=0$	**M1**	**1.1**	3 term quadratic in $\tan\theta$ obtained following attempt at use of Pythagoras; may be implied if equation correct, otherwise substitution must be seen; allow one coefficient error
		$(3\tan\theta-1)(\tan\theta+1)[=0]$	**B1**	**2.1**	value(s) for $\tan\theta$ obtained from **their** quadratic by valid method; must see correct factorisation or correct use of formula **oe**
		$\tan\theta=-1$, $\tan\theta=\frac{1}{3}$	**A1**	**1.1**	both correct
		$-45°, 135°$, 18.4° to 18.435°, $-161.6°$ to $-161.565°$ or $-162°$	**A1**	**1.1**	any two values correct; dependent on award of previous **A1**
			A1	**3.2a**	all four values correct with no extras in the range; ignore values outside the range if **A0A0** allow **SC1** for $-\frac{\pi}{4}, \frac{3\pi}{4}$, 0.32175 to 0.322, -2.8198 to -2.82 **NB B0M1B0A1A1A1** is possible
		Alternative method 1 $3+2\sin\theta\cos\theta-4\cos^2\theta=0$			from multiplying through by $\cos^2\theta$
		$3+\sin 2\theta-4(\frac{1}{2}(\cos 2\theta+1))[=0]$	**B1**		substitution of correct double angle formulae in correct equation
		$\sqrt{5}\sin(2\theta-63.4)=-1$ or $\sqrt{5}\cos(2\theta+63.4)=1$	**M1**		$R\sin(2\theta\pm\alpha)=k$ *or* $R\cos(2\theta\pm\alpha)=k$
		$\sin(2\theta-63.4)=-\frac{1}{\sqrt{5}}$ or $\cos(2\theta+26.6)=\frac{1}{\sqrt{5}}$	**M1** **A1**		value for $\sin(2\theta-63.4)=-\frac{1}{\sqrt{5}}$ or $\cos(2\theta+26.6)=\frac{1}{\sqrt{5}}$ obtained by valid method; **A1** if all correct

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		$-45°, 135°$, 18.4° to 18.435°, $-161.6°$ to $-161.565°$ or $-162°$	**A1**		any two values correct; dependent on award of previous A**1**
			A1		all four values correct with no extras in the range; ignore values outside the range if **A0A0** allow **SC1** for $-\frac{\pi}{4}, \frac{3\pi}{4}$, 0.32175 to 0.322, -2.8198 to -2.82
		Alternative method 2 $(2\sin\theta\cos\theta)^2 = (4\cos^2\theta - 3)^2$			from multiplying through by $\cos^2\theta$ and squaring both sides
		$4\sin^2\theta - 4\sin^4\theta = (1 - 4\sin^2\theta)^2$ **oe** **or** $4\cos^2\theta - 4\cos^4\theta = (4\cos^2\theta - 3)^2$ **oe**	**B1**		correct substitution of $\cos^2\theta = 1 - \sin^2\theta$ or $\sin^2\theta = 1 - \cos^2\theta$ in correct equation
		$20\sin^4\theta - 12\sin^2\theta + 1 = 0$ **or** $20\cos^4\theta - 28\cos^2\theta + 9 = 0$	**M1**		3 term quadratic in $\sin^2\theta$ or $\cos^2\theta$ obtained following use of Pythagoras – may be implied if equation correct, otherwise substitution must be seen; allow one coefficient error
		$\sin^2\theta = 0.5, 0.1$ **or** $\cos^2\theta = 0.5, 0.9$	**B1**		values for $\sin^2\theta$ or $\cos^2\theta$ obtained from **their** quadratic by valid method; must see correct factorisation or correct use of formula
			A1		both values of $\sin^2\theta$ or $\cos^2\theta$ correct
		$-45°, 135°$, 18.4° to 18.435°, $-161.6°$ to $-161.565°$ or $-162°$	**A1**		any two values correct; dependent on award of previous **A1**
			A1		all four values correct with no extras in the range; ignore values outside the range if **A0A0** allow **SC1** for $-\frac{\pi}{4}, \frac{3\pi}{4}$, 0.32175 to 0.322, -2.8198 to -2.82 **NB B0M1B0A1A1A1** is possible
			[6]		